

Single-Cell Cerebellum Annotation Analysis

1. Single-Cell Annotation

The raw cerebellum.h5ad dataset containing 4,962 cells × 14,782 genes was processed for single-cell annotation. The workflow included preprocessing, normalization, dimensionality reduction, and UMAP generation. The annotated AnnData dataset contains the normalized expression matrix, UMAP embeddings, and final_cell_type labels for downstream classifier analysis.

Cell Type / Subpopulation	Number of Cells	Percentage
Bergmann Glia	1,943	39.16%
Vascular Endothelial Cells	1,321	26.62%
Granule Neurons	1,317	26.54%
Oligodendrocytes	381	7.68%
Total annotated cells	4,962	100%

Key Marker Gene Signatures

Cell Type	Key Marker Gene Signatures
Bergmann Glia	SLC1A3 (GLAST), HOPX, FABP7, S100B
Vascular Endothelial Cells	CLDN5, FLT1, PECAM1, BSG
Granule Neurons	FAT2, GABRA6, RBFOX3 (NeuN), GRIN2C
Oligodendrocytes	MBP, PLP1, MOG, MAG, CNP

2. Preprocessing and Dimensionality Reduction

The workflow used Scanpy for preprocessing. If adata.X was unavailable, the counts layer was used when available, followed by the normalized layer as a fallback. The expression matrix was normalized to a target sum of 10,000 counts per cell and log1p-transformed.

The top 2,000 highly variable genes were selected. PCA was performed using 30 components, followed by construction of a 15-neighbor graph using 30 PCs. UMAP was then generated.

3. Generated Analysis Files

File	Description
annotated_cerebellum.h5ad	Annotated AnnData dataset with normalized expression matrix, UMAP embeddings, and final_cell_type labels
annotate_cerebellum.py	Standalone Python cell annotation workflow
cerebellum_cell_distribution.csv	Cell-type abundance table
cerebellum_differential_markers.csv	Top 50 ranked marker genes per cell type with log2FC and adjusted p-values
cerebellum.h5ad	Raw input dataset: 4,962 cells × 14,782 genes

4. Summary

The cerebellum single-cell dataset containing 4,962 cells was processed and categorized into four reported cell populations: Bergmann Glia, Vascular Endothelial Cells, Granule Neurons, and Oligodendrocytes. Bergmann Glia represented the largest population at 39.16%, followed by Vascular Endothelial Cells at 26.62% and Granule Neurons at 26.54%. The annotated h5ad dataset and supporting annotation, abundance, and differential-marker files were generated for downstream analysis.

Note: Classifier benchmarking results and the classifier identity were not provided in the supplied information.